

Stochastic optimal control

- Optimal TSB: a forward-backward SDE system

Enables learning!!!

Forward: $dX_t = dY_t + Z_t dt, X_0 \sim \rho_0$

Backward: $dX_t = dY_t - \tilde{Z}_t dt, X_1 \sim \rho_1$

Reference:

$$dY_t = -cLY_t dt + g_t dW_t$$

TSB-learning model

$$Z_t \approx Z_t(\theta) \quad l(x_0; \phi)$$

$$\tilde{Z}_t \approx \tilde{Z}_t(\phi) \quad l(x_1; \theta)$$

Learnable Trainable

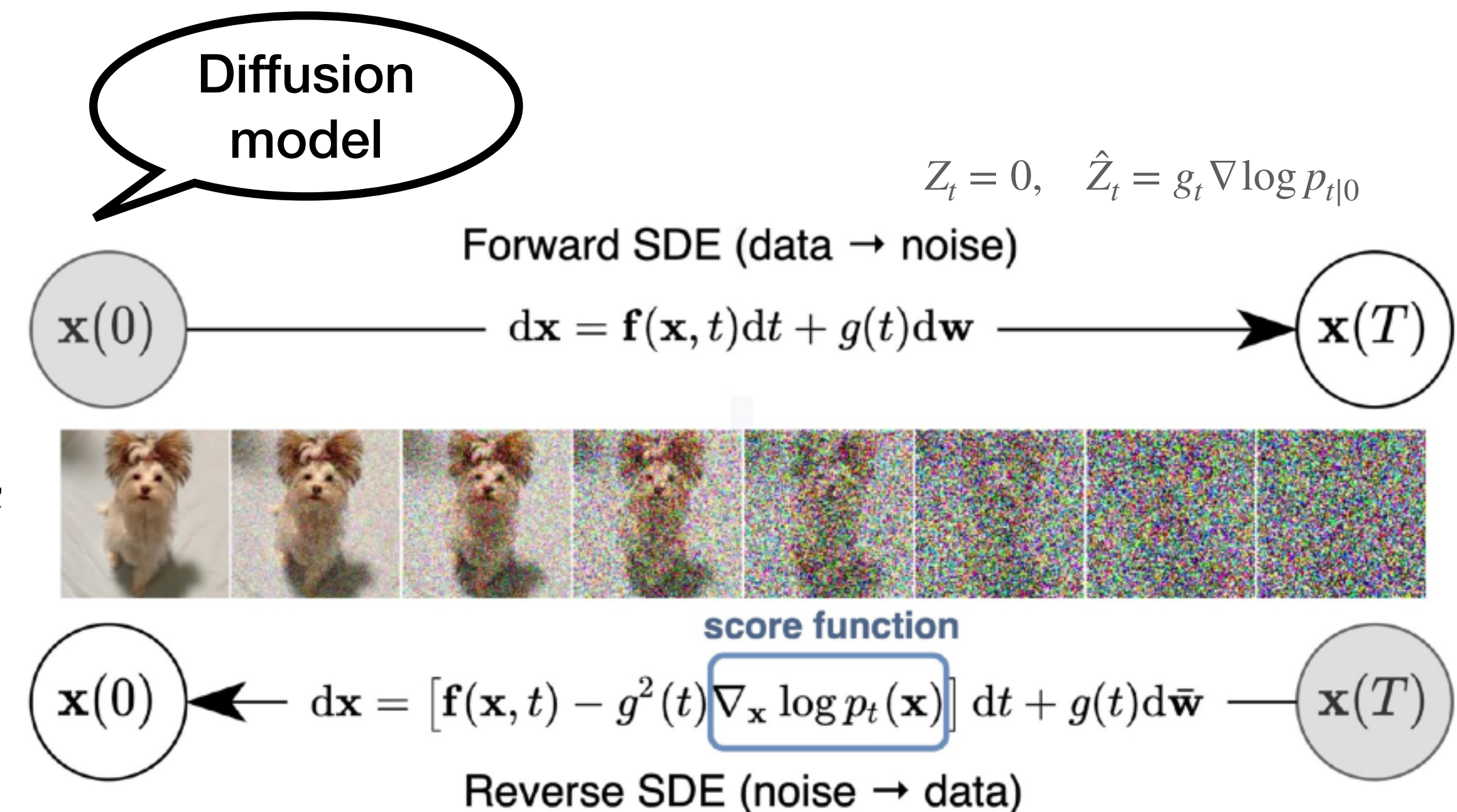
- Learnable models $(Z_t(\theta), \hat{Z}_t(\hat{\theta}))$ for optimal policies $(Z_t, \hat{Z}_t)$

- NNs, GNNs, simplicial NNs

- Trainable objective: max likelihood

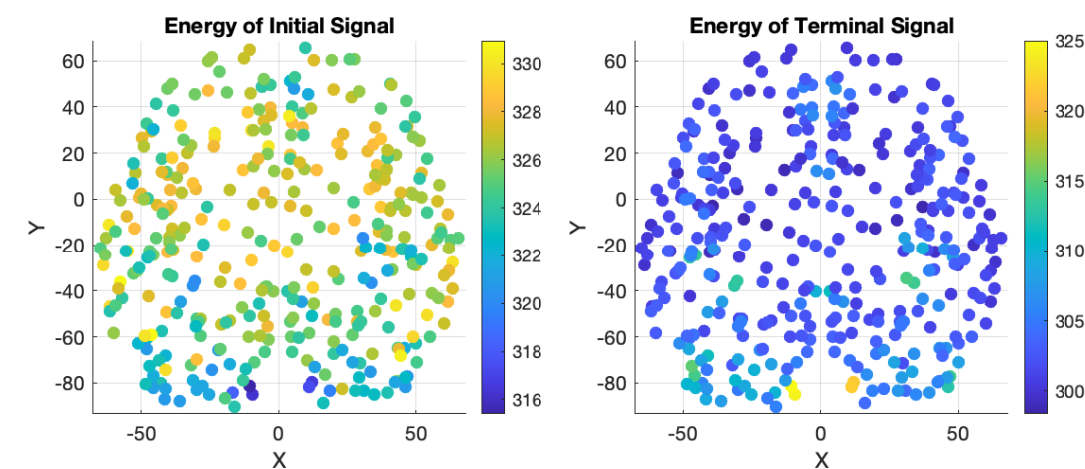
$$\mathcal{L}_{TSB}(x_0) = \mathbb{E} [\log \nu_1(X_1)] - \int_0^1 \mathbb{E} \left[\frac{1}{2} \|Z_t\|^2 + \frac{1}{2} \|\hat{Z}_t\|^2 + \nabla \cdot (g_t \hat{Z}_t - f_t) + \hat{Z}_t^\top Z_t \right] dt$$

- Unifies diffusion model



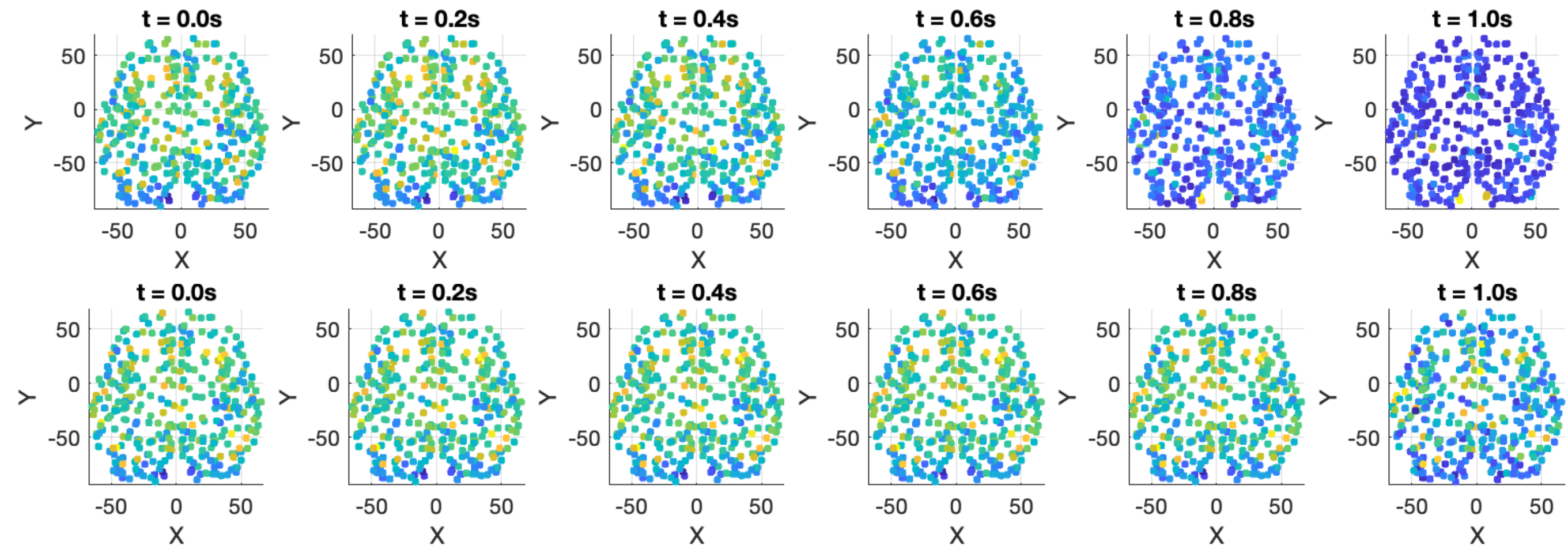
Topological signal matching

- Brain fMRI signals



liberal

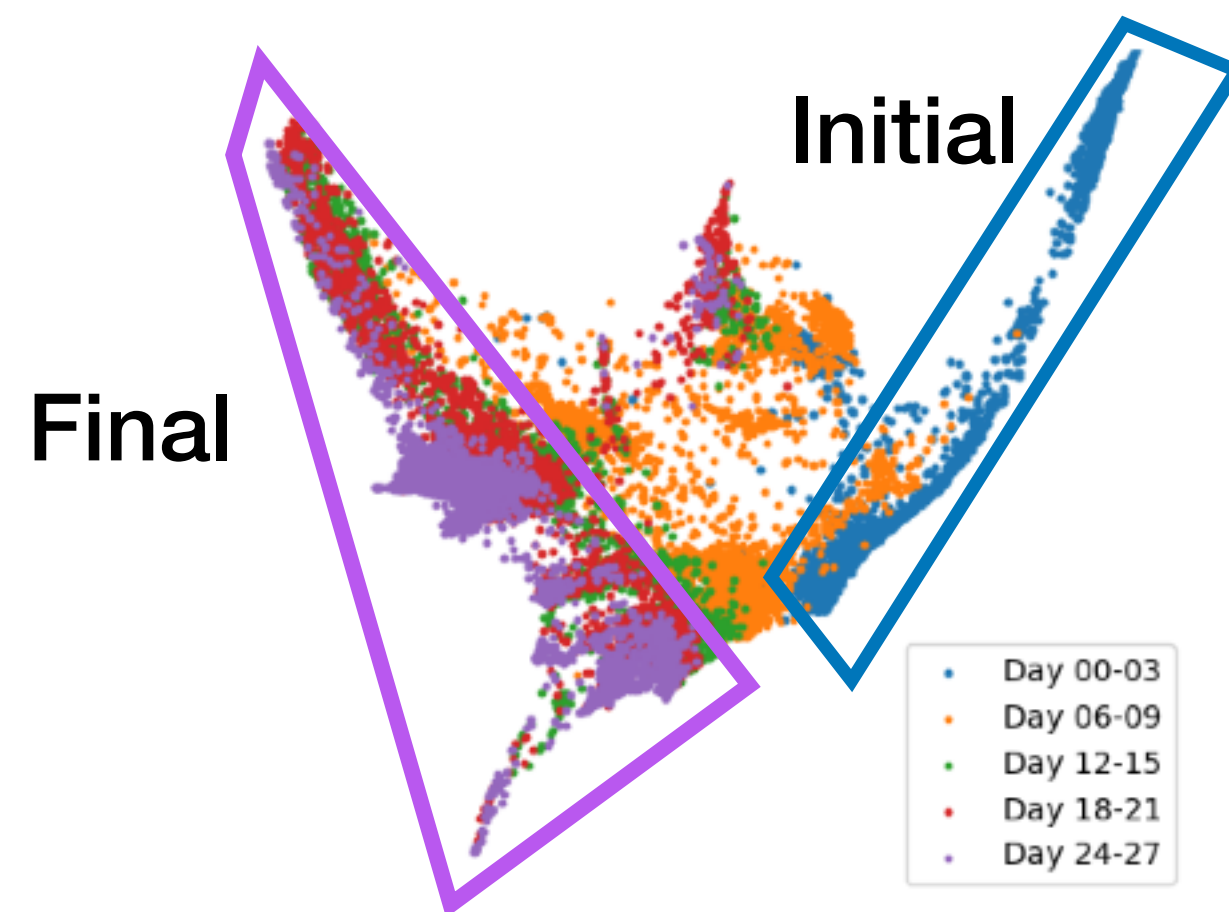
aligned



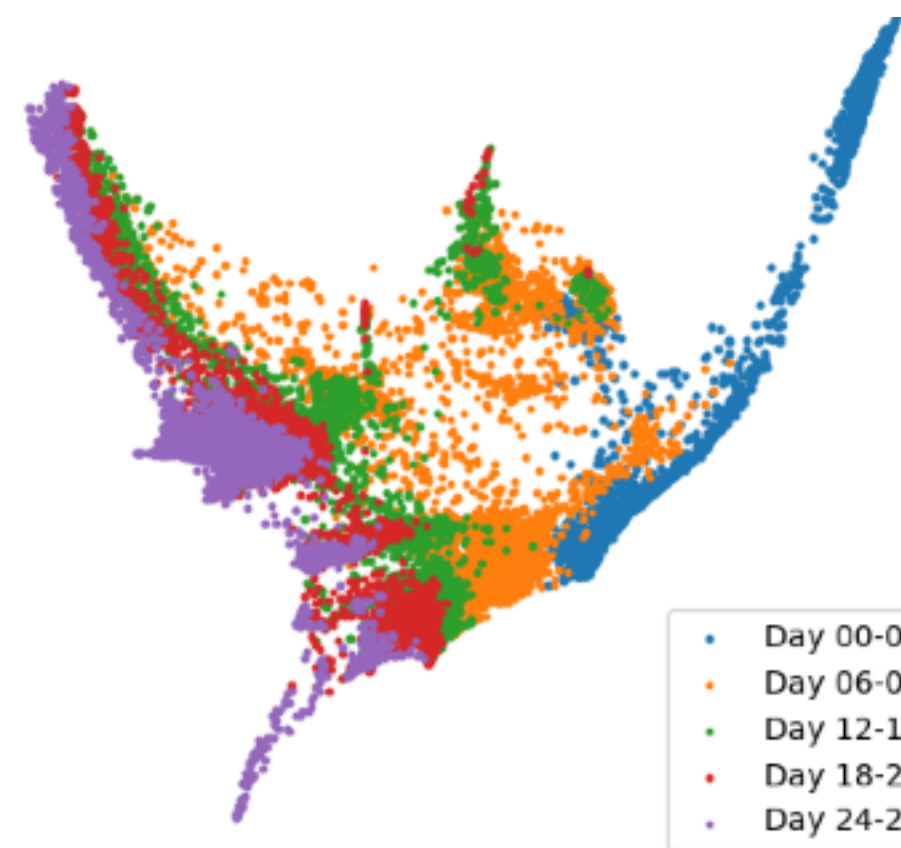
TSB-VE

SB-VE

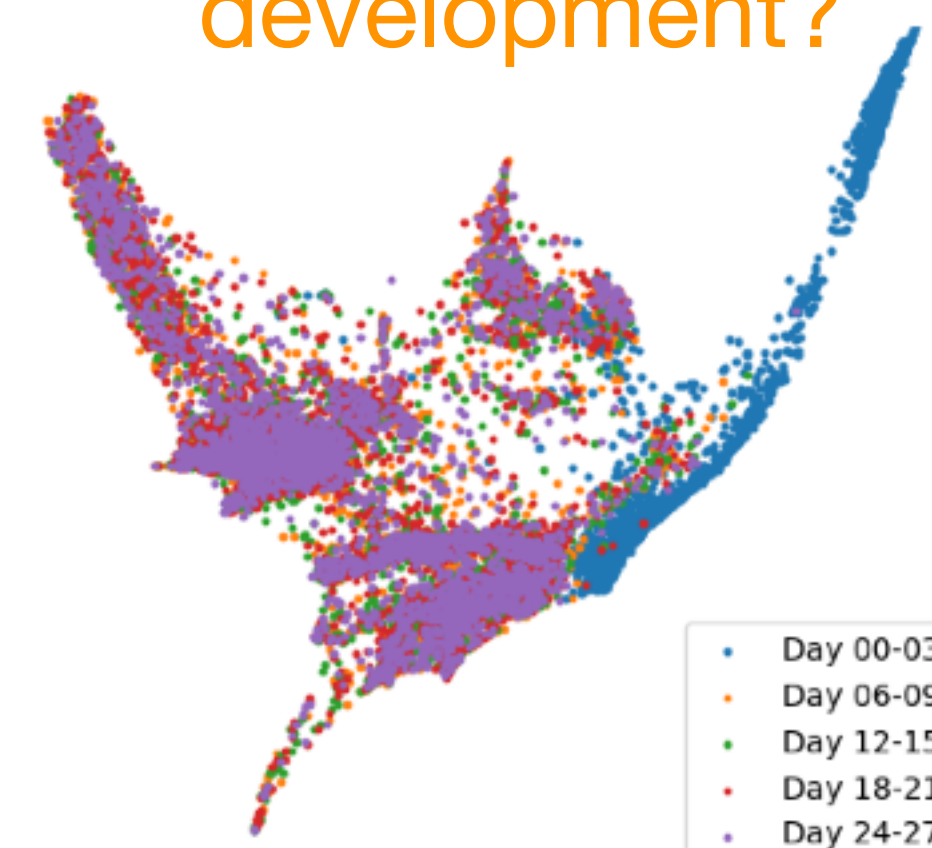
- Single-cell dynamics [Moon 2019, Tong 2023, 2024]



Groundtruth



TSB-BM



SB-BM

Potentials for chronic disease development?